



Medical Microbiology

Lecture Fourteen

Virology

Third stage- College of Dentistry

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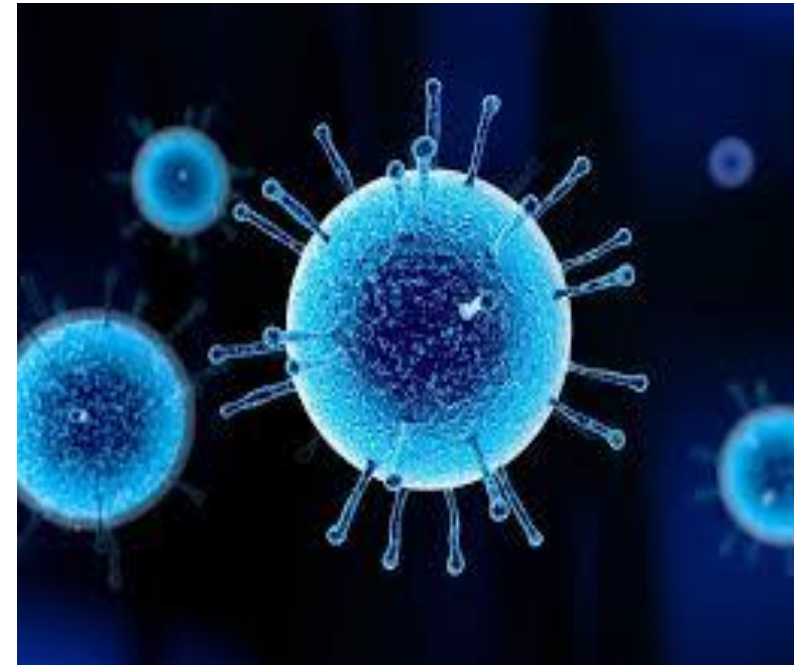
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What is a Virus?

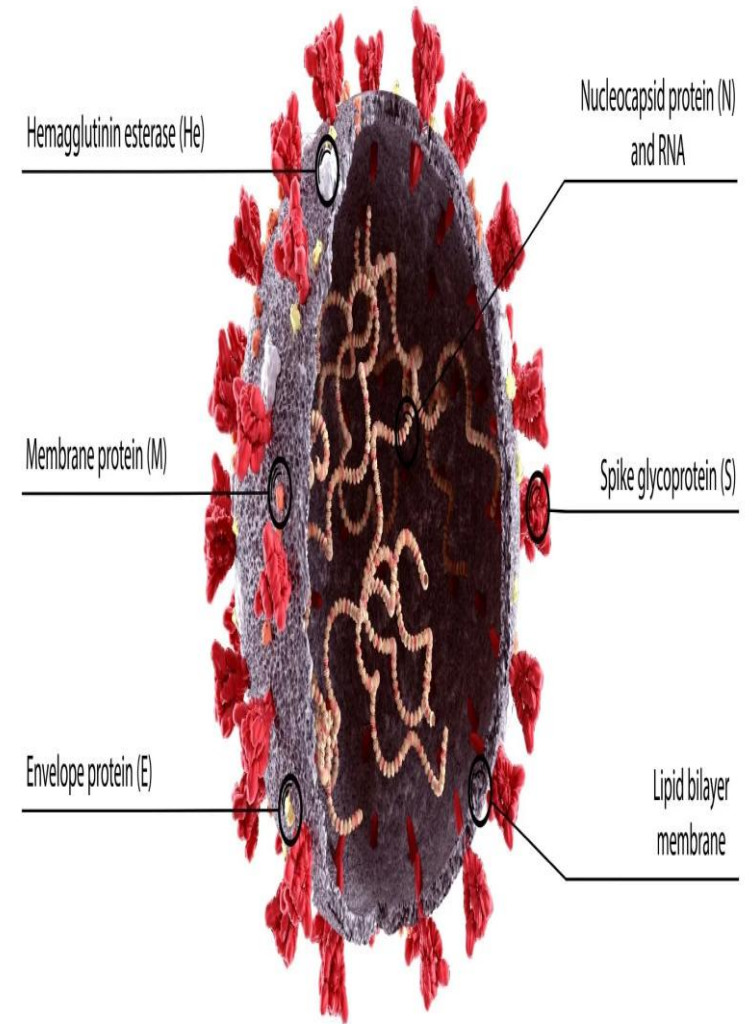
- A virus is an acellular infectious agent that requires a host cell to replicate, It consists of **DNA or RNA** surrounded by a protein coat.
- Lacks metabolic machinery and depends on host cells for energy and protein synthesis.

Basic Structure of a Virus:

- **Viral Genome:** The genetic material of a virus, which may be **DNA or RNA**, single-stranded or double-stranded.



- **Capsid:** The **protein coat** that surrounds and protects the viral genetic material.
- **Capsomere:** The **structural protein subunits** that make up the capsid.
- **Envelope (Optional):** A lipid bilayer derived from the host membrane, often with **glycoproteins** that aid in **protection, attachment and interact with host cells.**
- **Spikes:** **Glycoprotein projections** on the viral surface that help the virus attach to host cells.



Viral Replication Cycle:

1- Attachment (Adsorption):

- Virus binds to specific **host cell receptors**.
- Example: HIV binds to **CD4 receptors** on T cells.

2- Penetration (Entry):

- **Endocytosis** (e.g., Influenza) or **Membrane fusion** (e.g., HIV).
- Inject genetic material into cells.

3- Uncoating:

- Viral genome is released into the host cytoplasm or nucleus.

4- Replication and Transcription:

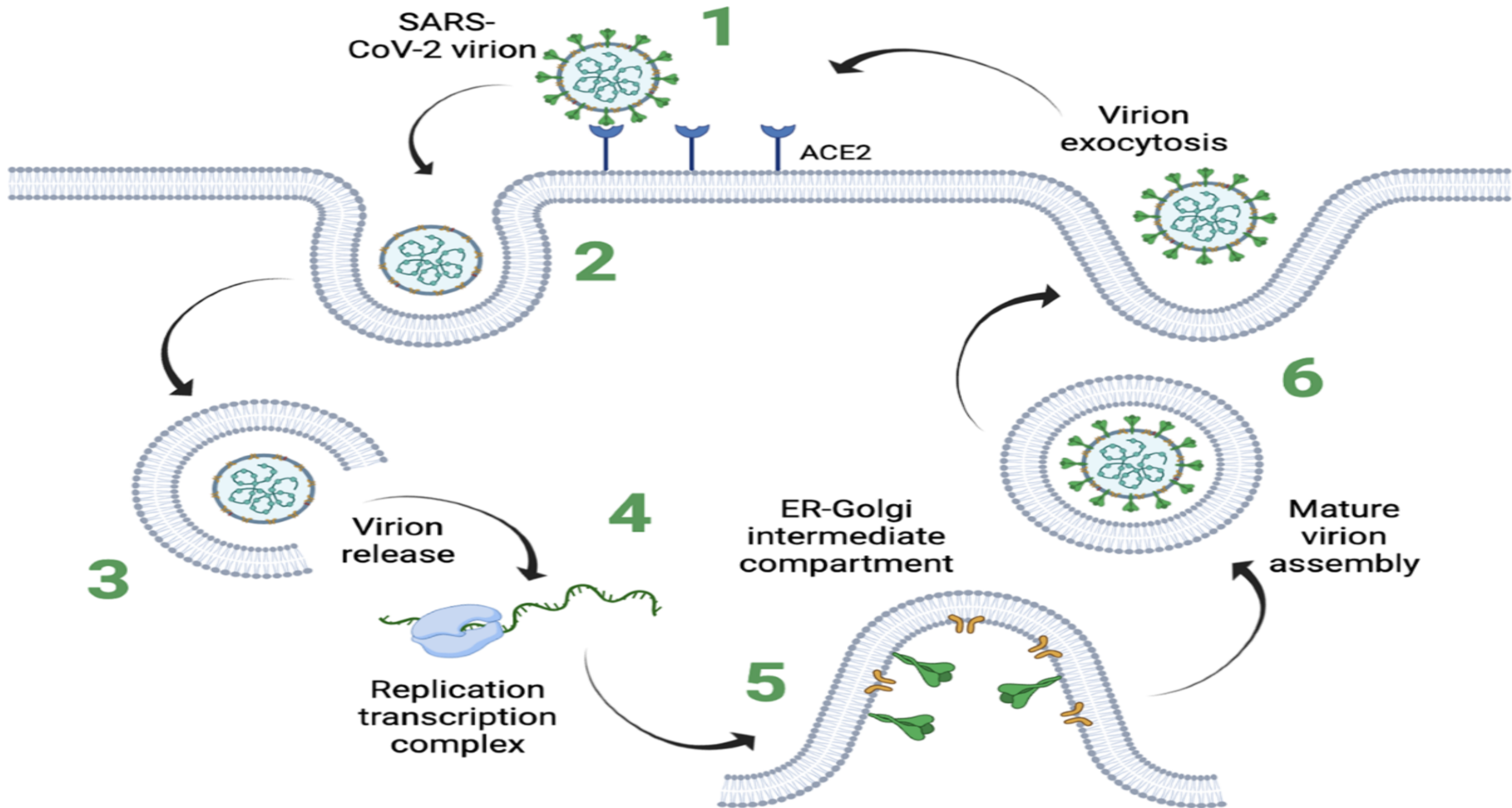
- DNA viruses usually replicate in the **nucleus** (except poxviruses).
- RNA viruses replicate in the **cytoplasm**.
- **Note:** Retroviruses (e.g., HIV) use **reverse transcriptase** to integrate into the host genome.

5- Assembly and Maturation:

- New viral particles are formed.

6- Release:

- **Lysis** (cell bursting) in non-enveloped viruses.
- **Budding** in enveloped viruses



Pathogenesis of Viral Infections:

Mechanisms of Viral Disease

1-Direct Cell Damage: Lysis of infected cells (e.g., Poliovirus).

2-Immune Response: Cytokine storm in severe infections (e.g., COVID-19).

3-Oncogenesis: Some viruses cause cancer (e.g., HPV → cervical cancer, EBV → Burkett's lymphoma).

4-Latency: Viruses remain dormant (e.g., Herpes simplex, Varicella-zoster).

Types of Viral Infections:

- **Acute infections:** Rapid onset, short duration (e.g., Influenza).
- **Chronic infections:** Persistent viral shedding (e.g., Hepatitis B, C).
- **Latent infections:** Virus remains dormant and reactivates later (e.g., Herpes virus).
- **Slow infections:** Long incubation period, progressive damage (e.g., HIV).

Host Immune Response to Viruses:

Innate Immunity:

- 1-Interferons (IFN- α , IFN- β , IFN- γ , IFN- λ): Inhibit viral replication.
- 2-Natural Killer (NK) Cells: Destroy infected cells.
- 3-Macrophages: Phagocytosis of virus particles.

Adaptive Immunity:

1-Humoral Immunity (B cells & Antibodies):

- Antibodies prevent viral entry (e.g., IgA in mucosal immunity).

2-Cell-Mediated Immunity (T Cells):

- CD8+ Cytotoxic T Cells kill infected cells.

Laboratory Diagnosis of Viral Infections:

1- Direct Detection:

- **PCR (Polymerase Chain Reaction):** Detects viral nucleic acids (e.g., COVID-19 testing).
- **Electron Microscopy:** Identifies viral morphology.
- **Antigen Detection:** Rapid tests (e.g., Influenza antigen test).

2- Serological Testing:

- **ELISA:** Detects antibodies (IgM, IgG) against viruses.
- **Western Blot:** Used for HIV confirmation.

3- Cell Culture:

- Some viruses grow in cultured cells (e.g., Herpes viruses).

Antiviral Therapy and Prevention:

1- Antiviral Drugs: Inhibit Viral Replication:

- Acyclovir: Used for Herpes simplex.,
- Oseltamivir (Tamiflu): Inhibits Influenza neuraminidase.
- Highly Active Antiretroviral Therapy (HAART): Used for HIV.

2- Vaccines:

- Live Attenuated Vaccines: Strong, long-lasting immunity (e.g., Measles, Mumps, Rubella).
- Inactivated Vaccines: Safer for immunocompromised patients (e.g., Hepatitis A).
- mRNA Vaccines: New technology used for COVID-19 (Pfizer, Moderna).

Medically Important Viruses

Virus	Disease	Mode of Transmission
Influenza virus	Flu	Respiratory droplets
Hepatitis B virus (HBV)	Hepatitis B	Blood, sexual contact
Human Immunodeficiency Virus (HIV)	AIDS	Blood, sexual contact
SARS-CoV-2	COVID-19	Respiratory droplets, aerosols
Human Papillomavirus (HPV)	Cervical cancer	Sexual contact
Herpes Simplex Virus (HSV)	Cold sores, genital herpes	Direct contact

Emerging and Re-Emerging Viruses:

- **SARS-CoV-2 (COVID-19)**
- **Ebola virus**
- **Zika virus**
- **H5N1 Avian Influenza**

Factors Contributing to Viral Emergence:

- Increased travel.
- Climate change and zoonotic spillovers.
- Antiviral resistance.

Conclusion:

- Viruses are obligate intracellular pathogens that **require a host** for replication.
- The immune system plays a crucial role in **viral defense**.
- **Diagnosis** relies on molecular tests, serology, and cell culture.
- **Prevention** through vaccination and antiviral drugs is essential in controlling viral diseases.



Any Question?



Thank you